

JEE MAIN REVISION MASTER: STATISTICS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 124 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Mean, Variance & Standard Deviation	119 PYQs	Dispersion
High	Measures of Central Tendency & Median	13 PYQs	Central Tendency

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Mean, Variance & Standard Deviation	$\bar{x} = \frac{\sum x_i}{n}, \quad \sigma^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2, \quad \sigma = \sqrt{\text{Variance}}, \quad \text{CV} = \frac{\sigma}{\bar{x}} \times 100$
Measures of Central Tendency & Median	$\text{Empirical: } \text{Mode} \approx 3 \times \text{Median} - 2 \times \text{Mean}, \quad \text{Median} = l + \left(\frac{N/2 - CF}{f} \right) h$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Mean, Variance & Standard Deviation (Very High Yield — 119 PYQs)

Core Principle: Measures of central tendency, variance calculation using mean of squares minus square of mean, and dispersion scaling.

■ **Caution / JEE Pitfall:** Adding a constant c to every observation changes the mean ($\bar{x}' = \bar{x} + c$), but leaves the VARIANCE completely UNCHANGED ($\sigma'^2 = \sigma^2$).

■ **Speed Trick:** If every observation is multiplied by k : new mean = $k\bar{x}$, new standard deviation = $|k|\sigma$, and new variance = $k^2\sigma^2$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A set of four observations has mean 1 and variance 13. Another set of six observations has mean 2 and variance 1. Then, the variance of all..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The mean deviation about the mean for the data x_i & 5 & 7 & 9 & 10 & 12 & 15 & f_i & 8 & 6 & 2 & 2 & 2 & 6" [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "If the mean of the data Class 5-10 10-15 15-20 20-25 25-30 30-35 Frequency 2 k 28 54 $k + 1$ 5 is 21, then k is one of the roots of the equa..." [Ans: N/A]

3.2 Measures of Central Tendency & Median (High Yield — 13 PYQs)

Core Principle: Computation of median and mode for discrete and continuous distributions, and Pearson empirical relationship.

■ **Caution / JEE Pitfall:** Sum of absolute deviations is minimum when taken about the MEDIAN ($\sum |x_i - M| = \min$), whereas sum of squared deviations is minimum about the MEAN ($\sum (x_i - \bar{x})^2 = \min$).

■ **Speed Trick:** Mean is heavily sensitive to extreme outlier values, while Median is robust against outliers.

Representative Source PYQs:

- **[2026 Morning Shift] Q#3 (MCQ):** "Suppose that the mean and median of the non-negative numbers 21, 8, 17, a, 51, 103, b, 13, 67, (a > b), are 40 and 21, respectively. If the mean deviation about the median is 10, then the value of a is" **[Ans: D]**
- **[2026 Evening Shift] Q#3 (MCQ):** "If the mean deviation about the median of the numbers k, 2k, 3k, ..., 1000k is 500, then the value of k is" **[Ans: N/A]**
- **[2026 Evening Shift] Q#4 (MCQ):** "A variable X takes values 0, 0, 2, 6, 12, 20, ..., n(n-1) with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, {}^nC_4, {}^nC_5, \dots$ " **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

The mean deviation about the mean for the data $x_{(i)}$ 5 7 9 10 12 15 $f_{(i)}$ 8 6 2 2 2 6 is equal to:

- (A) (40 / 13)
- (B) (42 / 13)
- (C) (44 / 13) **[Correct]**
- (D) (46 / 13)

Correct Answer: C

Solution Analysis: For a discrete frequency distribution, the mean deviation about the mean is M.D. about mean = $\frac{\sum f_{(i)} |x_{(i)} - \bar{x}|}{\sum f_{(i)}}$ First, find the mean $\bar{x}$. Step 1: Calculate $\sum f_{(i)}$ and $\sum f_{(i)} x_{(i)}$
 $\sum f_{(i)} = 8 + 6 + 2 + 2 + 2 + 6 = 26$ and $\sum f_{(i)} x_{(i)} = 40 + 42 + 18 + 20 + 24 + 90 = 234$
 So, the mean is $\bar{x} = \frac{\sum f_{(i)} x_{(i)}}{\sum f_{(i)}} = \frac{234}{26} = 9$
 Step 2: Calculate $\sum f_{(i)} |x_{(i)} - \bar{x}|$ Since $\bar{x} = 9$, w

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Evening Shift (Q#4)

A variable X takes values 0, 0, 2, 6, 12, 20, ..., n(n-1) with frequencies

${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, {}^nC_4, {}^nC_5, \dots, {}^nC_n$, respectively. If the mean of this data is 60, then its median is :

- (A) 56 **[Correct]**
- (B) 42
- (C) 72
- (D) 90

Correct Answer: A

Solution Analysis: The given variable X takes values 0, 0, 2, 6, 12, 20, ..., n(n-1) with corresponding frequencies ${}^nC_0, {}^nC_1, {}^nC_2, {}^nC_3, {}^nC_4, {}^nC_5, \dots, {}^nC_n$. We notice that the values of X follow the pattern $X = r(r-1)$, $r = 0, 1, 2, \dots, n$ and the corresponding frequency is nC_r . So this is a discrete frequency distribution with: variate value $x_r = r(r-1)$ frequency $f_r = {}^nC_r$. Step 1: Use the formula for mean. Mean is given by $\bar{X} = \frac{\sum f_r x_r}{\sum f_r}$. So, $\bar{X} = \frac{\sum {}^nC_r r(r-1)}{\sum {}^nC_r} = 0$. Given mean = 60, so